

Math 54 Discussion Worksheet

Name: _____

ID: _____

Problem 1

Using only real differentiation, prove that

$$\frac{d}{dt}e^{(\alpha+i\beta)t} = (\alpha + i\beta)e^{(\alpha+i\beta)t}.$$

It will help to use the identity

$$e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos(\beta t) + i \sin(\beta t)).$$

Problem 2

Solve

$$y'' - 3y' + 7y = t^4 e^t$$

Problem 3

Show that

$$y'' + \lambda^2 y = \sin(t), \quad y(0) = 0, y(\pi) = 1$$

has a solution iff $\lambda \neq \pm 1, \pm 2, \dots$